

Instructor: Date:

Learner: Regional Branch:

Program: Center:

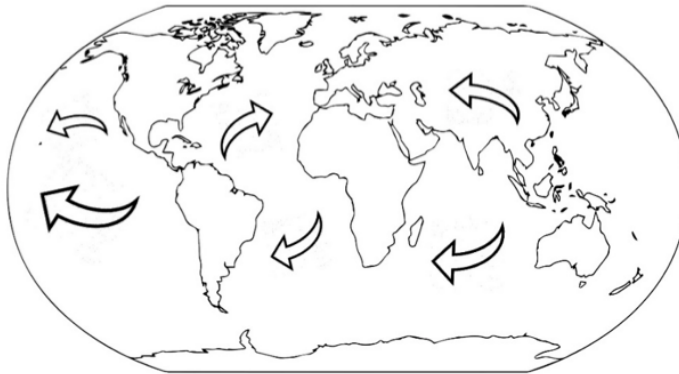
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Let's Start!

Arrows to Relate Classes



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can list the different kinds of association in UML.
- I can mention basic characteristics for different kinds of association in UML.
- I can describe how to show different kinds of associations with arrows.

Ask your partner:

- Why do you think it is necessary to have classes related?
- What do you think you need to do when associating classes?





Let's Explore!

A programmer describes the associations we generally find in UML*!

*"UML" stands for Unified Modelling Language



Instructions: Read the text below.

Programmer: When you want to use two classes, you need to associate them so they can work together, but this depends on what kind of association you want to present using UML diagrams.

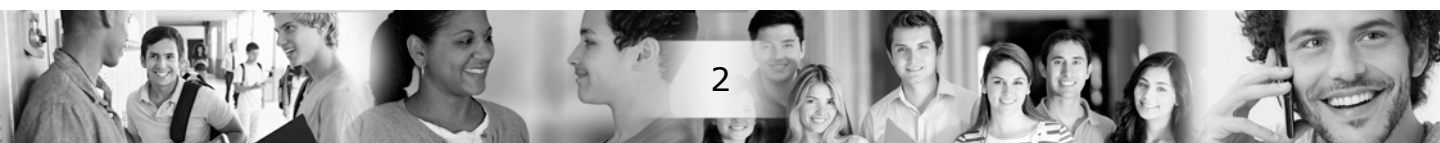
Assistant: How can I represent those associations, then?

Programmer: When you want to show different kinds of associations, you use arrows; plus, if you don't want to confuse the reader, you need to use the right arrow types.

Assistant: Okay! I want to denote my classes with arrows so I can make an association. What type do I need?

Programmer: If you don't want to make a strict association, you can sometimes use dependency. Here, class A depends on what class B defines. If you want to change something in class B, this affects class A; that's because class B has a static method. For this type of association, you use a dotted-line. When you don't want to depend on class B, you can usually use simple associations. In other words, you can use any object from class B because class A can generally have a class B attribute. A simple association uses a common arrow that represents just one direction. If you want to use a two-way direction, you can usually use the bidirectional association for common references between class A and class B when one depends on the other.

Now, when you want class B to have objects that belong to class A, you can usually use aggregation, so classes A and B have access to each other. Designers can sometimes think there is no distinction between aggregation and simple association but using a different arrow. Finally, if you want to use a similar aggregation process but you want to make more emphasis on the object of class B that belongs to class A, you can use composition by using a similar aggregation arrow.





Let's Practice!

According to the association you want to show, you can use a different kind of arrow.

Instructions: Read the sentences. Type in the missing words. Use the following Adverbs of Frequency: sometimes, always, never. Practice saying the sentences aloud. Check your pronunciation with your instructor.

Programmers can _____ show the different kinds of associations in their UML diagrams. For that purpose, programmers can _____ forget to use arrows. There are different kinds of associations with arrows; for example, programmers can _____ use a dotted-line to show dependency, or a one-direction arrow to show a simple association.

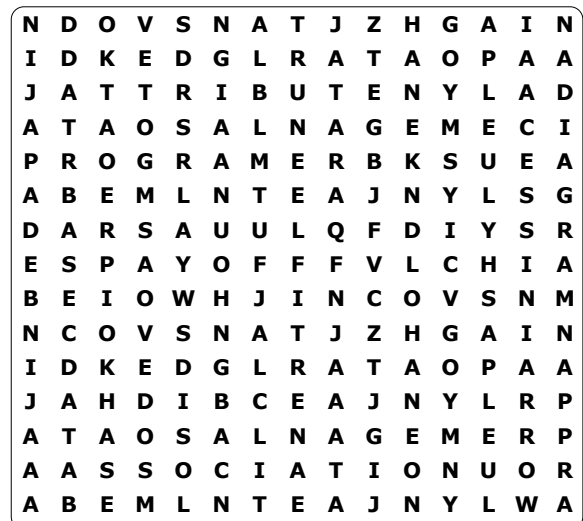


Let's Practice!

There are some types of associations that arrows can denote. Look for them in this Word Search.

Instructions: Find the words in the Word Search below.

1. You can always check the different types of associations that we use in our UML _____.
2. We can always _____ two different classes in UML.
3. In simple associations, we can sometimes use a common _____ that represents one direction.
4. A class A can generally have a class B _____.
5. A professional _____ can sometimes have problems deciding which type of association to include in UML.





Extension Activity

Let's create a poster of arrows!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about graphics that different kinds of associations use UML.
- Create a poster to explain the arrows used for different kinds of associations.
- Include examples and images.
- Share the poster with the rest of the class.

Self-Assessment

Yes

No

- I read different kinds of associations in UML. ☐ Yes ☐ No
- I associate different kinds of associations in UML with their respective arrows. ☐ Yes ☐ No
- I draw arrows for different kinds of associations in UML and I share the poster with the rest of the class. ☐ Yes ☐ No



Target Words

- dotted-line** (noun): a line you make using dots
e.g. For this type of association, you use a dotted-line.
- bidirectional** (adjective): something that goes in two opposite directions
e.g. If you want to use a two-way direction, you can usually use the bidirectional association for common references between class A and class B when one depends on the other.
- aggregation** (noun): the objects you collect from one group to another
e.g. Now, when you want class B to have objects that belong to class A, you can usually use aggregation, so classes A and B have access to each other.
- composition** (noun): the way you join objects together
e.g. Finally, if you want to use a similar aggregation process but you want to make more emphasis on the object of class B that belongs to class A, you can use composition by using a similar aggregation arrow.
- denote** (verb): to show something
e.g. Okay! I want to denote my classes with arrows so I can make an association.
- arrow** (noun): a symbol you use to give a specific direction to something
e.g. Okay! I want to denote my classes with arrows so I can make an association.



Learning Points

Verbs with Infinitives: Want to, in Affirmative and Negative Statements

You use **want to** plus a verb in its infinitive form to express wishes or desires.

You can use **want to** in affirmative statements to express there is an action you wish to perform. For affirmative sentences follow the simple present structure. Let's review it:

Subject + want to* + verb in base form + complement.

*Be careful! Don't forget to add a final "-s" to the verb when the subject of the sentence is a third person.

For example:

*She **wants to** learn more about the associations in UML.*

*They **want to** design UML diagrams.*

Now, you can also use **want to** in negative statement by following the simple present structure in its negative form:

Subject + auxiliary (does not / do not) + want to* + verb in base form + complement.

*Remember that when we use the auxiliary **does not** or **do not**, we do not conjugate the verb.

For example:

*She **doesn't want to** learn more about the associations in UML.*

*They **do not want to** design UML diagrams.*

Examples

- *When you want to show different kinds of association, you use arrows; if you don't want to confuse the reader, you need to use the right arrow types.*





Learning Points

Arrows to Denote Types of Association

In UML, you use **arrows** to denote types of association between classes.

The types of associations that arrows can denote are the following:

TYPE OF ASSOCIATION	EXPLANATION
Dependency	Two or more classes depend on each other.
Simple Association	Classes are associated, but they do not depend on each other.
Bidirectional Association	Dependable classes can affect one another.
Aggregation	There is a belonging relationship between two different classes.
Composition	Stronger than aggregation. It denotes that A is an elemental part of class B.